Hyeonsu Jeong

Curriculum Vitae

Ph.D. Candidate
School of Electrical Engineering, KAIST
291 Daehak-ro, Yuseong-gu, Daejeon, Korea

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Education

Korea Advanced Institute of Science and Technology — Daejeon, South Korea
Integrated *M.S.* and *Ph.D.* Program in Electrical Engineering, GPA: 4.2/4.3 — 2021 – 2026 (expected)
Advisor: Hye Won Chung

Korea Advanced Institute of Science and Technology — Daejeon, South Korea
B.S. in Electrical Engineering, Cum Laude, GPA: 3.86/4.3 — 2016 – 2021

Gyeonggi Science High School for the Gifted — Suwon, South Korea
2013 – 2016

Research Interests

Information Theory and Applications to Machine Learning

Keywords

- Crowdsourcing and Data Efficiency
- Knowledge Distillation and Self-Distillation
- Robust Learning and Label Correction
- Deep Learning Theory and Analysis
- Probability and Statistics to Machine Learning

Research Experience

Inference and Information for Data Science Lab (IIDS Lab at KAIST) — 2021 – 2026 (expected)
Advisor: Prof. Hye Won Chung

Koh Young Technology, Academic Internship, KAIST Co-op Program — 2018 – 2019
- Study on anomaly detection in production processes
- Study on the graph structure of production process management systems

Robot Intelligence Technology Lab (RIT Lab at KAIST) — 2018
Advisor: Prof. Jong Hwan Kim

Publications

Rethinking Self-Distillation: Label Averaging and Enhanced Soft Label Refinement with Partial Labels
Hyeonsu Jeong, Hye Won Chung
International Conference on Learning Representations (ICLR), 2025

Recovering Top-Two Answers and Confusion Probability in Multi-Choice Crowdsourcing
Hyeonsu Jeong, Hye Won Chung
International Conference on Machine Learning (ICML), 2023

Patents

Method and Device for Recovering Top-Two Answers and Confusion Probability in Multi-Choice Crowdsourcing
Hye Won Chung, Hyeonsu Jeong.
Patent Application No. KR 10-2023-0074573.

Method and Electronic Apparatus for Displaying Inspection Result of Board
Jong Myoung Lee, Duk Young Lee, Suhyeong Choi, Hyeonsu Jeong.
Patent Application No. KR 10-2019-0153629, US 16/624,619, CN 201980003048.9, EP 19817118.3

Electronic Device and Method for Determining Cause of Mounting Defect in Components Mounted on Substrate
Jong Myoung Lee, Duk Young Lee, Suhyeong Choi, Hyeonsu Jeong.
Patent Application No. KR 10-2019-0077004, US 16/621,142, CN 201980002555.0, EP 19812895.1

Projects

National Research Foundation (Basic Research Lab) 2024 – Present
Title: Developing Reliable Foundation Models with Theoretical Framework and Scalable Personalization

Inst. for Information and Communications Technology Promotion 2024 – Present
Title: Non-invasive Near-Infrared based AI Technology for the Diagnosis of Brain Diseases

National Research Foundation (Excellent Early-Career Research Award) 2021 – 2024
Title: Machine learning utilizing incomplete data

Inst. for Information and Communications Technology Promotion 2022
Title: Ensuring high AI learning performance with only a small amount of training data

National Research Foundation (AI-Data Science Strategic Research Funding) 2021
Title: Value-centered big data acquisition and processing

Teaching Assistant

Programming Structure for Electrical Engineering (Course Code: 35.209) Fall 2023, Spring 2024

Probability and Introductory Random Processes (Course Code: 35.210) Spring 2022, 2023; Fall 2023

Introduction to Information Theory and Coding (Course Code: 35.326) Spring 2024

Information Theory (Course Code: 35.623) Fall 2021, 2022, 2024

Skills

Languages: Korean (Native), English (Intermediate)

Computer Languages: Python, LaTeX, C/C++, MATLAB